



Integrated Design. Engineered Performance.

CAPABILITIES & SERVICES PORTFOLIO

Integrated Construction, Engineering & Building Performance

A coordinated suite of construction, building systems, commissioning, energy, and comprehensive structural engineering services.

CONSTRUCTION | ENGINEERING | COMMISSIONING | ENERGY

From initial assessment to verified performance.

INTEGRATED DELIVERY

One Team. One Coordinated Delivery Strategy.

SCG connects construction, engineering, building systems, commissioning and energy expertise into a single service platform built around project outcomes.

Our combined capabilities support new construction, renovation, modernization and performance improvement - from early assessment and design through execution, verification and ongoing optimization.

A Full Project Lifecycle

1. Assess & Plan - Define existing conditions, priorities, risks, energy opportunities, scope, schedule and delivery strategy.
2. Design & Coordinate - Align architectural intent with structural, mechanical, electrical and plumbing requirements while addressing constructability and code.
3. Build & Modernize - Execute commercial, residential, interior and site work with coordinated labor, materials, subcontractors, quality and schedule management.
4. Verify & Commission - Test and document that systems, assemblies and building performance meet project requirements and design intent.
5. Optimize & Sustain - Reduce operating cost, extend asset life, document results and support energy compliance, incentives and long-term performance.

Combined Service Portfolio

- General Contracting & Construction Management - Commercial, residential, renovation and project-delivery services.
- Commercial Interiors & Tenant Fit-Outs - Coordinated architectural, MEP, life-safety and finish scopes.
- MEP & Facility Modernization - Design coordination, systems integration and targeted equipment upgrades.
- Commissioning & Quality Assurance - Functional testing, deficiency resolution and documented conformance.
- Energy & Compliance - Audits, modeling, retrofits, M&V, BEPS support, rebates and tax incentives.
- Structural Engineering - Delegated design, new-building teaming, temporary works, verification, BIM and reality-capture workflows.

CONSTRUCTION SERVICES

General Contracting & Commercial Interiors

End-to-end delivery for commercial construction, renovation and tenant-specific build-outs.

General Contracting & Project Delivery

- General Contracting - Residential & Commercial - Coordinate labor, materials, subcontractors, schedule and execution from mobilization through completion.
- Construction & Project Management - Plan and oversee preconstruction, procurement, budget, quality, safety, schedule, design requirements and closeout.
- Trade & Vendor Coordination - Sequence field activities, manage interfaces and maintain clear responsibility for deliverables.
- Quality, Safety & Documentation - Track inspections, submittals, deficiencies, corrective actions, turnover records and owner requirements.

Interior Renovations & Tenant Fit-Outs

- Metal Stud Framing - Interior wall framing and partition systems.
- Drywall - Installation, finishing, repair and coordinated backing for walls and ceilings.
- Acoustical Ceiling Tile - Suspended acoustical ceiling systems and finishes.
- Painting & Interior Finishes - Coordinated finish work for renovated and new spaces.
- Flooring - Installation and replacement across commercial interior applications.
- Kitchens & Bathrooms - Commercial and residential upgrades, reconfiguration and modernization.
- Doors, Glass & Millwork - Interior doors, conference-room glass, casework, architectural woodwork and finish carpentry.
- Electrical - High & Low Voltage - Integrated electrical scope for fit-outs and renovations.
- Fire Alarm Systems - Fire alarm installation, modification and project coordination.

CONSTRUCTION SERVICES

Residential Construction & Site Improvements

Comprehensive home construction, renovation and exterior improvement capabilities.

Residential Construction & Renovation

- Foundations - Foundation construction and related structural scope.
- Framing - Building framing for new work, additions and renovations.
- Windows, Roofing & Gutters - Envelope installation, replacement and coordinated repair.
- Flooring - Flooring installation and replacement across residential applications.
- Kitchen Renovations - Functional, aesthetic and performance upgrades.
- Bathroom Renovations - Reconfiguration and modernization of bathroom spaces.
- Basement Renovations - Build-outs and upgrades that expand usable living space.
- HVAC, Plumbing & Electrical - Building-system construction and upgrades coordinated with renovation work.
- Painting & Millwork - Interior/exterior painting, finish carpentry and architectural woodwork.

Exterior & Site Improvements

- Decks & Patios - New construction, replacement and renovation.
- Concrete Walkways - Site concrete for circulation and exterior improvements.
- Pavers & Flagstone - Hardscape installation for walks, patios and outdoor areas.
- Landscaping - Landscape improvements coordinated with site construction.
- Irrigation Systems - Irrigation installation and site integration.

BUILDING SYSTEMS

MEP Systems, Life Safety & Facility Modernization

Coordinated building-system solutions designed to reduce conflicts, support reliable operation and align with design intent and code requirements.

MEP Design, Coordination & Systems Integration

- Mechanical, Electrical & Plumbing Coordination - Plan and align building systems so they work together efficiently and without avoidable field conflicts.
- Systems Integration - Connect controls, equipment and supporting infrastructure to deliver intended function and performance.
- Electrical - High & Low Voltage - Electrical scope for commercial, residential and renovation projects.
- HVAC & Plumbing - Coordinated mechanical and plumbing upgrades within construction and renovation work.
- Fire Alarm Systems - Life-safety system installation, modification and project coordination.
- Constructability & Clash Resolution - Resolve structural, architectural and building-system conflicts before they disrupt field installation.

Facility Upgrades & Modernization

- Existing-Building Evaluation - Identify deficiencies and performance gaps affecting operations, safety and asset life.
- System & Equipment Modernization - Upgrade aging systems and equipment to current operational and efficiency standards.
- Repair-vs.-Replacement Planning - Use condition, performance and remaining useful life to support capital decisions.
- Coordinated Implementation - Integrate construction, MEP, controls and quality requirements within a practical project plan.

The result: coordinated systems, clearer scope, fewer conflicts and a building that performs as intended.

BUILDING PERFORMANCE

Commissioning, Energy & Compliance

Verification and optimization services that improve reliability, efficiency, comfort and return on investment.

Commissioning & Quality Assurance

- Commissioning - Verify that building systems are designed, installed, tested and operating in accordance with project requirements and design intent.
- Recommissioning - Revisit previously commissioned systems to restore or improve performance.
- Retro-Commissioning - Apply the commissioning process to existing buildings that were not previously commissioned.
- Building Envelope Commissioning - Verify enclosure-related design, installation and performance requirements.
- Quality Assurance - Document conformance, testing, deficiency resolution and closeout across the project.

Energy & Facility Analysis

- Energy Audits - Analyze energy use and identify efficiency opportunities.
- Facility Condition Assessments - Evaluate building systems, condition, remaining useful life and repair-versus-replacement options.
- Benchmarking - Establish current energy performance and compare results over time.
- Energy Modeling - Predict performance, evaluate alternatives and estimate savings.

Implementation, Verification & Incentives

- Energy Retrofits & Efficiency Improvements - Implement targeted equipment, system and enclosure upgrades to reduce consumption while maintaining or improving performance.
- Measurement & Verification - Confirm actual results and savings following implementation.
- BEPS Compliance - Support performance assessment, planning, documentation and compliance activities.
- Energy Rebates & Tax Incentives - Manage applicable utility and government programs that can reduce upfront cost and improve project return.

STRUCTURAL ENGINEERING

Temporary Works, Excavation & Construction Support

Specialty engineering for temporary systems that protect workers, support construction, maintain stability and address complex field constraints.

Structural engineering capabilities are furnished in collaboration with Structural Optimal Solutions, LLC (SOS), as applicable to project scope.

Temporary Works & Access Systems

- Scaffolding & Access Systems - Analysis and 3D modeling for frame, system, tube-and-clamp and 180-ft+ cantilevered scaffolds around complex geometries.
- Supported & Suspended Platforms - Engineering for stable elevated work areas and task-specific suspended platforms.
- Temporary Enclosures & Overhead Protection - Protective enclosures, sidewalk sheds and scaffolding-based overhead protection.
- Mast Climbers, Hoists & Swing Stages - Slab-capacity review, lateral-tie connection design, outrigger-beam detailing and temporary support when required.
- Formwork Engineering - Vertical and horizontal concrete formwork design with hydrostatic and dynamic load analysis in accordance with ACI 347.

Excavation, Shoring & Stability

- Excavation Shoring & Earth Retention - Temporary soldier piles and lagging, sheet piling, tiebacks, rakers and internal cross-lot bracing for deep excavations.
- Sheathing & Trench Protection - Temporary timber and steel sheeting, trench boxes and perimeter earth-retention barriers.
- Building Support & Wall Bracing - Shoring for modifications, historic rehabilitation and demolition, plus tilt-up and masonry-wall bracing during erection.
- Existing-Structure Verification - Assess dynamic, operational and concentrated point loads against structural capacity.

STRUCTURAL ENGINEERING

Delegated Structural Design

Complete analysis, calculations, connection engineering and coordinated shop-drawing packages for specialty structural systems.

Building Frame, Facade & Enclosure Systems

- Cold-Formed Steel Framing - Delegated engineering and shop drawings for curtain-wall framing, load-bearing studs, joists, roof trusses, shear walls and dynamic lateral wind-load connections.
- Structural Steel Connections - High-LOD 3D connection modeling; moment, shear and braced-frame connection design; truss-node analysis; fabrication drawings and capacity verification.
- Masonry, Stone & Facade Attachments - Engineering for granite, marble, limestone, cast stone, thin-stone veneer, precast and UHPC panels, shelf angles, continuous lintels and heavy facade anchors.
- Glazing, Curtain Wall & Storefronts - Wind, seismic and dynamic-load calculations plus connection engineering for unitized curtain walls, structural glass fins, point-supported glass, storefronts and skylights.
- Precast & Prestressed Concrete - Design, handling and erection-stress calculations, embed design and connection detailing for wall panels, hollow-core plank, double-tees and spandrel beams.

Deliverables can be coordinated from structural concept through delegated calculations, high-detail modeling, shop drawings, fabrication graphics and final connection detailing.

STRUCTURAL ENGINEERING

Specialty Components, Supports & Verification

Engineering for the secondary structural systems and project-specific loads that often control constructability, safety and coordination.

Specialty Structural Components

- Open-Web Steel Joists & Metal Decking - Connection verification, dynamic-load checks, concentrated equipment-load distribution and layout engineering for joist girders, bridging and floor/roof deck diaphragms.
- Timber & Mass Timber Connections - Custom steel knife plates, timber-to-concrete and timber-to-steel interfaces, Glulam trusses, CLT panels and specialty fasteners.
- Stairs, Railings & Monumental Steel - Design and dynamic-load checks for monumental, egress and spiral stairs, industrial catwalks, glass balustrades and handrails.
- Miscellaneous Metals & Equipment Supports - MEP equipment platforms, rooftop frames, mechanical dunnage, suspended utility grids, pipe bridges and custom steel fabrications.
- Fall Protection & Safety Anchorages - Dynamic-impact verification for rooftop davit pedestals, window-washing monorails and fall-arrest anchorages.

Existing-Structure & Equipment Load Verification

- Slab & Deck Load Verification - Evaluate concrete slabs, composite decks and framing under dynamic construction equipment such as Bobcats, forklifts, pumps, cranes and demolition rigs.
- Load Path & Re-Shoring Strategy - Trace gravity and lateral loads, verify floor capacities and design underlying re-shoring grids when wheel or point loads exceed allowable limits.
- MEP Equipment Supports - Analyze and design structural supports for mechanical and other building equipment, including rooftop and suspended applications.

STRUCTURAL ENGINEERING

New Buildings, Modular & Industrial Systems

Primary structural design teaming and practical long-term structural solutions for conventional, modular and specialized facilities.

New Building Structural Design - Peer Partnership

- Concrete Structures - Reinforced concrete frames, post-tensioned and prestressed slabs, shear-wall lateral systems, and shallow or deep foundations.
- Steel Structures - Structural steel framing, composite floor decks, braced and moment-resisting frames, and complex long-span roof trusses.
- Wood & Timber Structures - Multi-story light-frame wood, heavy timber, CLT, mass-timber hybrid systems and engineered wood floor/roof assemblies.
- Masonry Structures - Engineering for masonry walls, cast stone, precast stone, stone attachments, lintels and related structural elements.
- Code & Standards Coordination - Design informed by applicable IBC, ASCE 7, ACI, AISC, NDS and jurisdictional requirements.

Advanced Modular Solutions

- Unitized Multi-Story Systems - Structural design for steel chassis framing, inter-module tie connections and stacked modular construction.
- Transport & Erection Analysis - Evaluate temporary stresses, lifting conditions and transportability.
- Full-Envelope Engineering - Stabilization, specialized multi-unit connections for clusters of up to five units, and weather-tight cladding and roofing systems.
- Multi-State Deployment - Adapt structural design to regional wind, snow and code requirements.

Specialized Industrial Systems

- Gantry & Rigging Design - Heavy-duty gantry runways and industrial sheds for work in protected, controlled environments.
- Complex Equipment Support - Project-specific frames, platforms and load paths for industrial and building equipment.

DIGITAL ENGINEERING

BIM, Finite Element Analysis & Reality Capture

Integrated digital workflows that improve coordination, analysis, constructability and documentation across project teams.

Software / Workflow	Applied Capability
Autodesk Navisworks Manage	Multi-discipline clash detection across structural, architectural, MEP, excavation/site and temporary-works models.
Autodesk Revit Structure	High-LOD structural models, reinforcement details, automated drawings, schedules and shared central files.
Robot Structural Analysis	3D finite-element analysis, dynamic wind/seismic calculations and code checking synchronized with Revit.
Advance Steel & AutoCAD	Steel connection details, cold-formed steel layouts, shop drawings and fabrication graphics.
Autodesk ReCap Pro	Laser-scan and aerial reality-capture integration for existing-condition modeling and retrofit planning.

Reality Capture & Advanced Technology

- Drone-Based Photogrammetry & Photometry - Upcoming - Planned UAV capabilities for aerial surveys, high-resolution 3D meshes, earthwork-volume tracking, visual site inspections and elevation mapping.
- Scan-to-BIM Workflows - Convert drone photogrammetry and laser scans into Revit environments for existing-capacity and retrofit-geometry evaluation.
- Material Optimization Modeling - Computational analysis to improve framing efficiency and reduce structural material quantities in concrete, steel and wood developments.

STRUCTURAL PARTNER PROFILE

Structural Optimal Solutions, LLC

A specialist structural engineering partner focused on delegated design, temporary works and new-building design teaming.

Founded in 2025 by Johnny Abudayeh, P.E., SOS brings more than 12 years of specialized engineering experience to delegated structural design, specialty temporary works and new-building design teaming. Its practice connects structural analysis, detailed engineering and field constructability from schematic concept through connection detailing.

Principal	Johnny Abudayeh, P.E., M.ASCE
Office	24 Courthouse Square #503, Rockville, MD 20850
Direct	(202) 669-5018
Email	johnny.a@sos-structural.com

Leadership & Technical Credentials

- 12+ Years of Experience - Structural engineering, delegated design, project management and system integration.
- Education - Master of Engineering in Civil Engineering with structural focus and Bachelor of Civil Engineering, University of Maryland, College Park; graduate focus included project management, team leadership and portfolio strategy.
- Professional Engineer Licensure - Maryland #46361; Virginia #64245; District of Columbia #PE40000705; North Carolina #61071; Pennsylvania #PE097686; Delaware #PE32005; New Jersey #24GE06272000.
- Business Registrations - Maryland #W25650268; Virginia foreign entity #11893382; District of Columbia foreign entity #C00008514095.
- Technical Standards - IBC, ASCE 7, ACI 318, AISC Steel Construction Manual, ACI 347, NDS and Maryland Building Performance Standards.

SOS combines structural design expertise in reinforced concrete, structural steel and wood/mass timber with BIM, finite-element analysis and constructability-focused delivery.

SELECTED EXPERIENCE

Signature Structural Past Performance

Representative experience demonstrating complex temporary works, access systems, rigging support, historic constraints and modular innovation.

Amazon Helix Mixed-Use Tower**Arlington, Virginia**

Collaborated with Clark Construction to engineer specialized access scaffolding and structural support platforms for a 350-ft spiral tower.

National Air & Space Museum**Washington, District of
Columbia**

Structural engineering for a 120-ft clear-span steel protection structure with integrated heavy-duty gantry runways supporting skylight-glass rigging.

New Jersey State Capitol Restoration**Trenton, New Jersey**

Engineered a 180-ft cantilevered scaffold system around the historic dome so masonry restoration could proceed without unapproved loads on historic masonry.

Scalable Modular Building Systems**Multi-Site**

Designed multi-unit modular structural systems with engineered chassis frames, interlocking unit connections and custom envelope attachments.

National Zoo Aviary Project**Washington, District of
Columbia**

Project recognized with an ABC Excellence in Construction Award.

CLIENT EXPERIENCE

Built Around Your Project

Every engagement is shaped around the building, operating priorities, project constraints and the outcome the client needs to achieve.

A Coordinated Four-Phase Approach

1. Discover - Understand existing conditions, project goals, operational constraints and success criteria.
2. Develop - Translate priorities into coordinated scope, design, schedule, budget and implementation strategy.
3. Deliver - Execute with disciplined project management, trade coordination, structural and MEP integration, quality assurance and communication.
4. Demonstrate - Test, verify, document and optimize performance so the completed project supports long-term value.

What SCG Brings Together

- Integrated Expertise - Construction, structural engineering, MEP, commissioning and energy.
- Lifecycle Perspective - Decisions informed by first cost, performance, maintainability, constructability and asset life.
- Coordinated Execution - Clear alignment among design, structural requirements, field work, testing and operational needs.
- Performance Verification - Documented quality and measured results where applicable.
- Project-Specific Solutions - A service mix tailored to scope, building type, jurisdiction and delivery method.

Ready to move your project forward? Contact Superior Construct Group to discuss the right combination of services for your goals.



Integrated Design. Engineered Performance.

Service availability and final scope are project-specific and subject to applicable licensing, permitting, jurisdictional, contractual and teaming requirements.